

Exploratory Data Analysis Project

Presented by:

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Dataset Overview

	Product_ID	Sale_Date	Sales_Rep	Region	Sales_Amount	Quantity_Sold	Product_Category	Unit_Cost	Unit_Price	Customer_Type	Discount
0	1052	03-02-2023	Bob	North	5053.97	18	Furniture	152.75	267.22	Returning	0.09
1	1093	21-04-2023	Bob	West	4384.02	17	Furniture	3816.39	4209.44	Returning	0.11
2	1015	21-09-2023	David	South	4631.23	30	Food	261.56	371.40	Returning	0.20
3	1072	24-08-2023	Bob	South	2167.94	39	Clothing	4330.03	4467.75	New	0.02

Discount	Payment_Method	Sales_Channel	Region_and_Sales_Rep
0.09	Cash	Online	North-Bob
0.11	Cash	Retail	West-Bob
0.20	Bank Transfer	Retail	South-David
0.02	Credit Card	Retail	South-Bob

The dataset
comprises of
1000 Columns
and **14 Rows**

Missing Values:
No missing or null values
were detected in the dataset;
all fields are complete and
usable for analysis

- **Datetime Columns:** Sale_Date (converted from DD-MM-YYYY format)
- **Identifiers:** Product_ID, Sales_Rep
- **Categorical Columns :** Region, Product_Category, Customer_Type, Payment_Method, Sales_Channel
- **Numeric Columns:** Quantity_Sold, Unit_Price, Unit_Cost, Sales_Amount, Discount

	count	mean	std	min	25%	50%	75%	max	skew	kurtosis
Product_ID	1000.0	1050.128000	29.573505	1001.000000	1024.000000	1051.000000	1075.000000	1100.000000	0.028362	-1.216665
Sales_Amount	1000.0	5019.265230	2846.790126	100.120000	2550.297500	5019.300000	7507.445000	9989.040000	0.004116	-1.206440
Quantity_Sold	1000.0	25.355000	14.159006	1.000000	13.000000	25.000000	38.000000	49.000000	0.001644	-1.201944
Unit_Cost	1000.0	2475.304550	1417.872546	60.280000	1238.380000	2467.235000	3702.865000	4995.300000	0.025323	-1.192784
Unit_Price	1000.0	2728.440120	1419.399839	167.120000	1509.085000	2696.400000	3957.970000	5442.150000	0.023338	-1.188520
Discount	1000.0	0.152390	0.087200	0.000000	0.080000	0.150000	0.230000	0.300000	-0.057089	-1.227610
Total_Revenue	1000.0	5019.265230	2846.790126	100.120000	2550.297500	5019.300000	7507.445000	9989.040000	0.004116	-1.206440
Total_Cost	1000.0	63842.093640	55398.598783	60.280000	18591.320000	48453.135000	97755.397500	235808.090000	0.970648	0.118308
Profit	1000.0	-58822.828410	55549.189117	-235588.270000	-91927.975000	-43401.315000	-14001.440000	8865.370000	-0.974140	0.134635
Profit_Margin	1000.0	-31.168042	102.461187	-1808.297146	-23.460517	-9.477552	-2.931847	0.990833	-11.178463	161.930722

Numeric Summary

★ Descriptive Statistics Summary (Numeric Features)

- Sales and pricing metrics show moderate variation: **Sales_Amount** avg ≈ ₹5019, **Unit_Price** ≈ ₹2728, **Quantity_Sold** avg ≈ 25 units.
- Cost values vary widely (**Total_Cost** min ≈ ₹60, max ≈ ₹235,808), reflecting diverse product types.
- Discounts remain generally low (**mean 0.15**), with occasional high-discount cases.
- Profitability is highly inconsistent: **Mean profit** ≈ -₹58,822, with large spread and extreme outliers.
- Profit Margin** shows strong skewness and very high kurtosis, indicating irregular distribution and many loss-making transactions.

Categorical Summary

Region_and_Sales_Rep		count
0	North-Eve	64
1	East-Bob	60
2	East-David	59
3	South-David	56
4	North-David	56
5	East-Eve	55
6	North-Charlie	55
7	West-Bob	54
8	East-Alice	52
9	West-David	51

Product_Category		count
0	Clothing	268
1	Furniture	260
2	Electronics	246
3	Food	226

Sales_Rep		count
0	David	222
1	Eve	209
2	Bob	208
3	Alice	192
4	Charlie	169

Region		count
0	North	267
1	East	263
2	West	244
3	South	226

Payment_Method		count
0	Credit Card	345
1	Bank Transfer	342
2	Cash	313

Customer_Type		count
0	New	504
1	Returning	496

Sales_Channel		count
0	Retail	512
1	Online	488

Data Cleaning & Preparation

- Converted *Sale_Date* to datetime format (DD-MM-YYYY)
- Converted numeric columns: Sales_Amount, Quantity_Sold, Unit_Cost, Unit_Price, Discount
- Removed whitespace from text fields
- Created new calculated fields:**
 - Total_Revenue = Sales_Amount
 - Total_Cost = Unit_Cost × Quantity_Sold
 - Profit = Total_Revenue – Total_Cost
 - Profit Margin = Profit / Total_Revenue
- Extracted time features: **Month, Month Number**
- Created **Discount Band** (0–1%, 1–5%, etc.)



Sales Performance Dashboard: Key Metrics

₹5.01M

Total Revenue

-₹58.8M

Total Profit

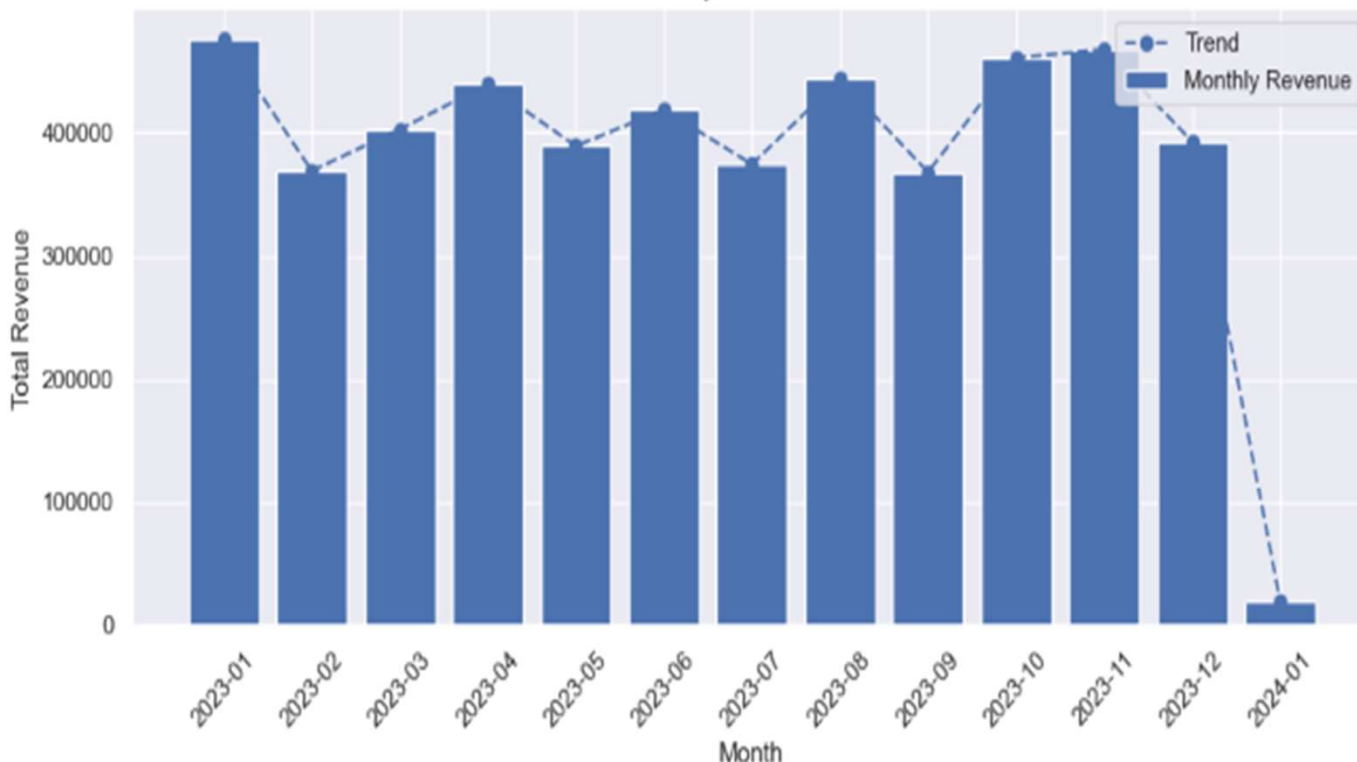
-31%

Average Profit
Margin

1,000

Total Transactions

Monthly Total Revenue



- ❖ Revenue remains stable with strong peaks in **Jan, Aug, Oct, and Nov**, indicating clear seasonal demand.
- ❖ Slight dips in **Feb, Jul, and Sep** suggest slower periods for targeted promotions.
- ❖ Overall trend shows **predictable seasonality**, supporting confident forecasting and inventory planning.

Monthly Revenue — Top 3 Product Categories



Monthly Revenue Trend

- Clothing, Furniture, and Electronics show **steady month-to-month demand**, with noticeable peaks in **January, April, June, and October**.
- Furniture displays the **highest revenue fluctuations**, while Clothing remains the **most stable and reliable performer**.
- Management Insight: **Stability in Clothing suggests it is a dependable revenue anchor**, while **targeted promotions during Furniture's peak months can maximize gains**.

Revenue by Region & Category

Sales by Region:

	Total_Revenue	Total_Profit
Region		
North	1369612.51	-15177330.14
East	1259792.93	-15167050.92
West	1235608.93	-14869867.95
South	1154250.86	-13608579.40

- **Clothing and Furniture** lead in revenue, making them core contributors to overall sales performance.

- Despite strong revenue, all categories show **heavy negative profits**, with Clothing having the largest loss.

- Focus should be on **cost reduction and pricing optimization**, especially in high-demand categories to turn sales volume into profitability.

- **North and East** generate the highest revenue, indicating strong demand and larger customer activity in these markets.

- All regions show **significant negative profit**, suggesting high operational or product costs across the board.

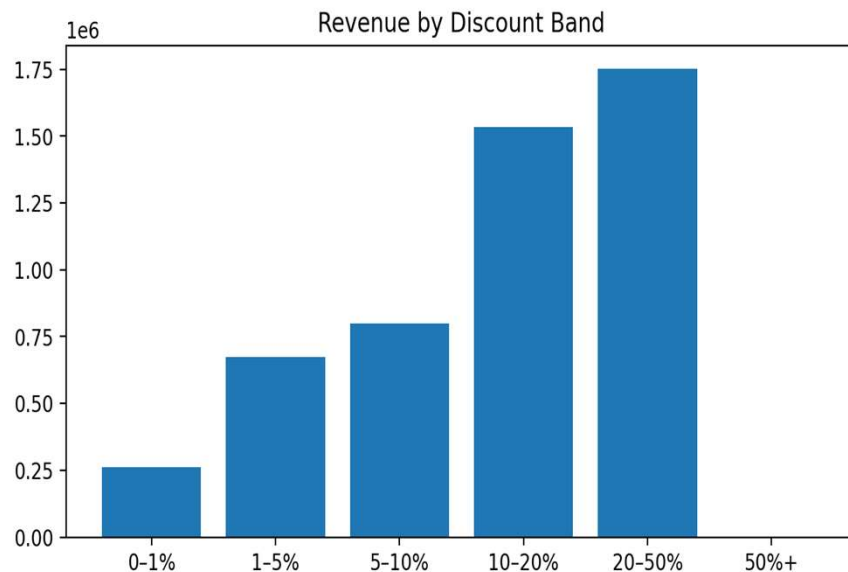
- Management should **audit cost drivers in each region**, prioritizing North and East for efficiency improvements due to their high sales volume.

Sales by Product Category:

	Total_Revenue	Total_Profit
Product_Category		
Clothing	1313474.36	-16260301.77
Furniture	1260517.69	-15290301.71
Electronics	1243499.64	-14753220.87
Food	1201773.54	-12519004.06

Revenue by Discount Band

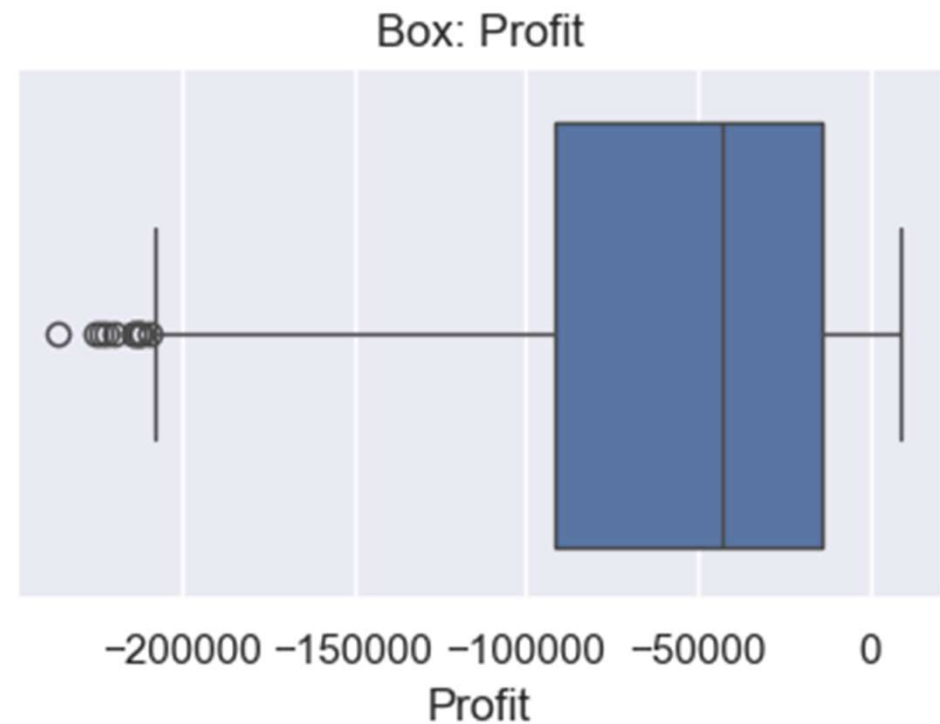
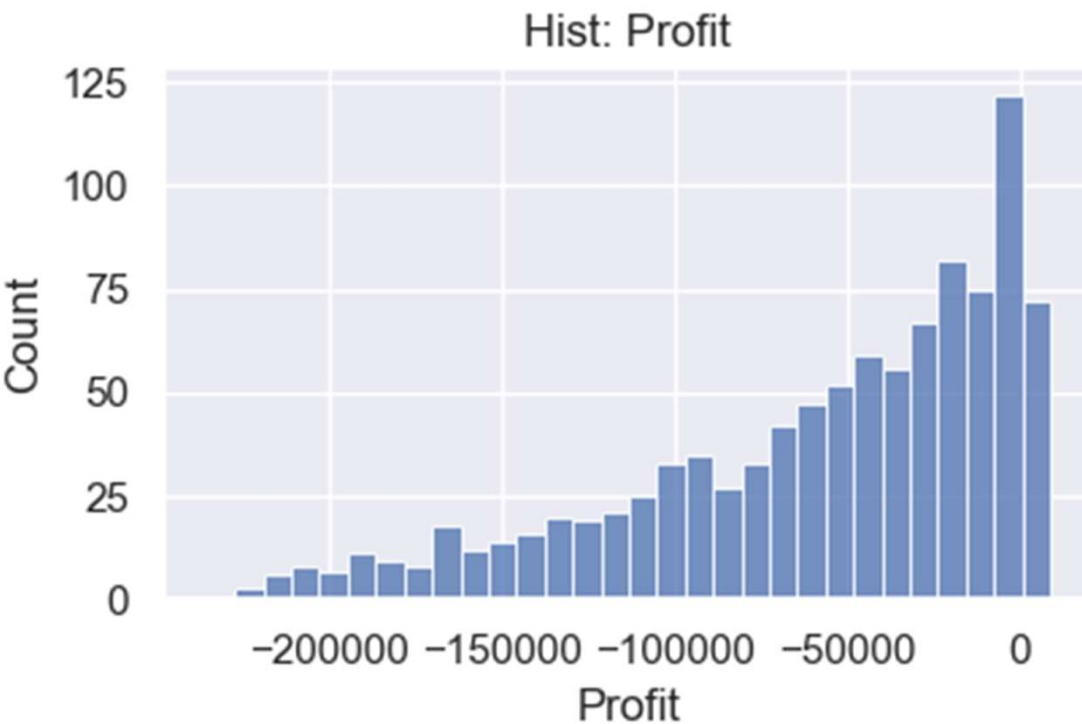
Discount_Band	Total_Revenue	Avg_Quantity	Transactions
0-1%	261007.66	25.943396	53
1-5%	673233.88	25.338583	127
5-10%	797818.84	26.658824	170
10-20%	1534754.11	24.517799	309
20-50%	1752450.74	25.378299	341
50%+	0.00	NaN	0



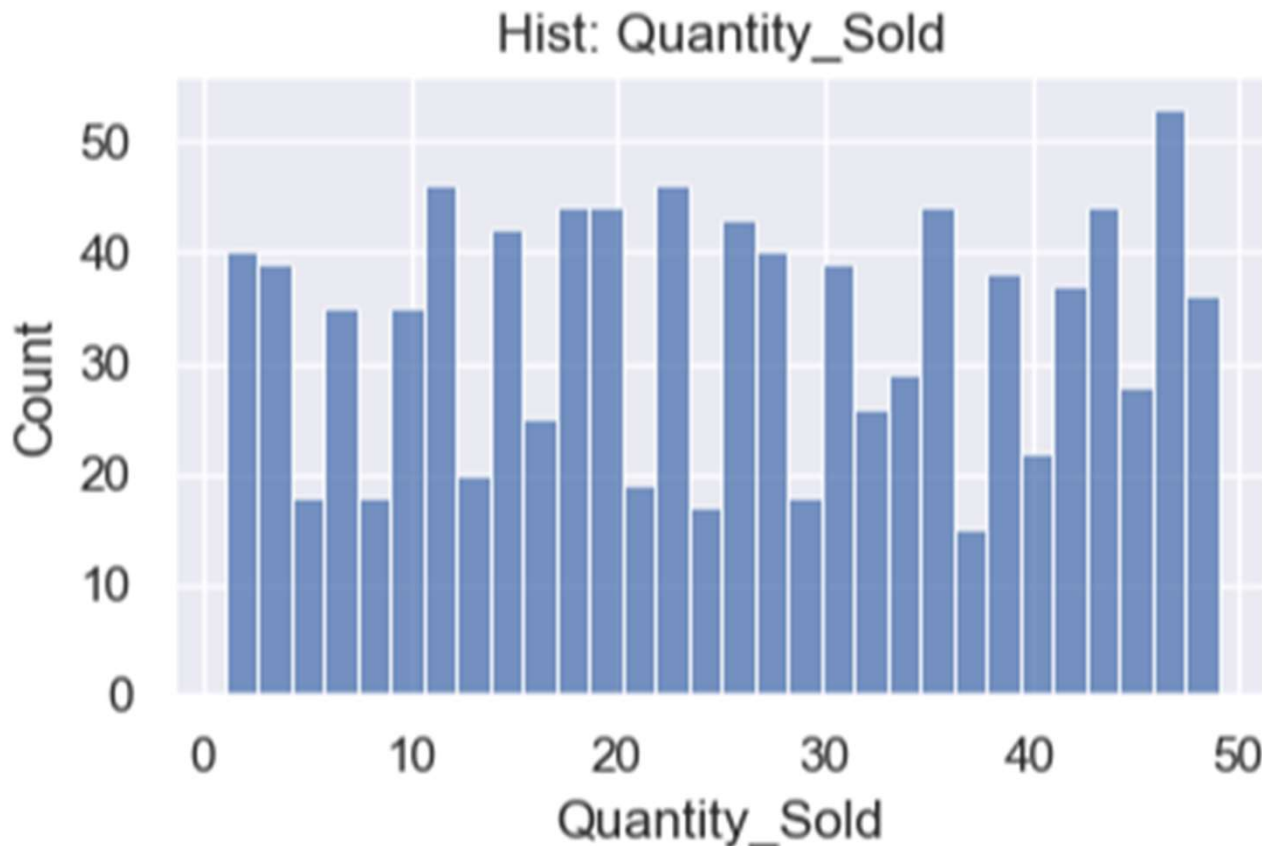
- Revenue **steadily increases** with higher discount levels, with the **20-50% band generating the maximum revenue**.
- The **20-50% range** also records the **highest transaction volume (341)**, showing strong customer response to significant discounts.
- Very low discount bands (0-5%) contribute the **least revenue**, indicating limited impact on buyer motivation.
- **Management Insight:** High discount tiers effectively boost both **sales volume and revenue** — useful for clearance, seasonal sales, or demand stimulation, but must be balanced against profitability.

Profit Distribution

- Profit values are heavily **skewed toward losses**, with most transactions falling between **–200K and –20K**, indicating consistently high cost-to-revenue ratios.
- The boxplot shows **significant negative outliers**, confirming operational inefficiencies or high-cost items driving losses.
- Management should **review pricing, cost structure, and discount strategy** to reduce loss-heavy transactions.



Quantity Sold Distribution



- **Inventory Planning Insight**

Consistent medium-range quantities indicate **predictable demand**, making stock planning easier and reducing the risk of overstock or shortages.

- **Customer Behavior Insight**

Limited bulk purchases show that customers mainly buy for **regular consumption**, not wholesale, helping refine targeting strategies.

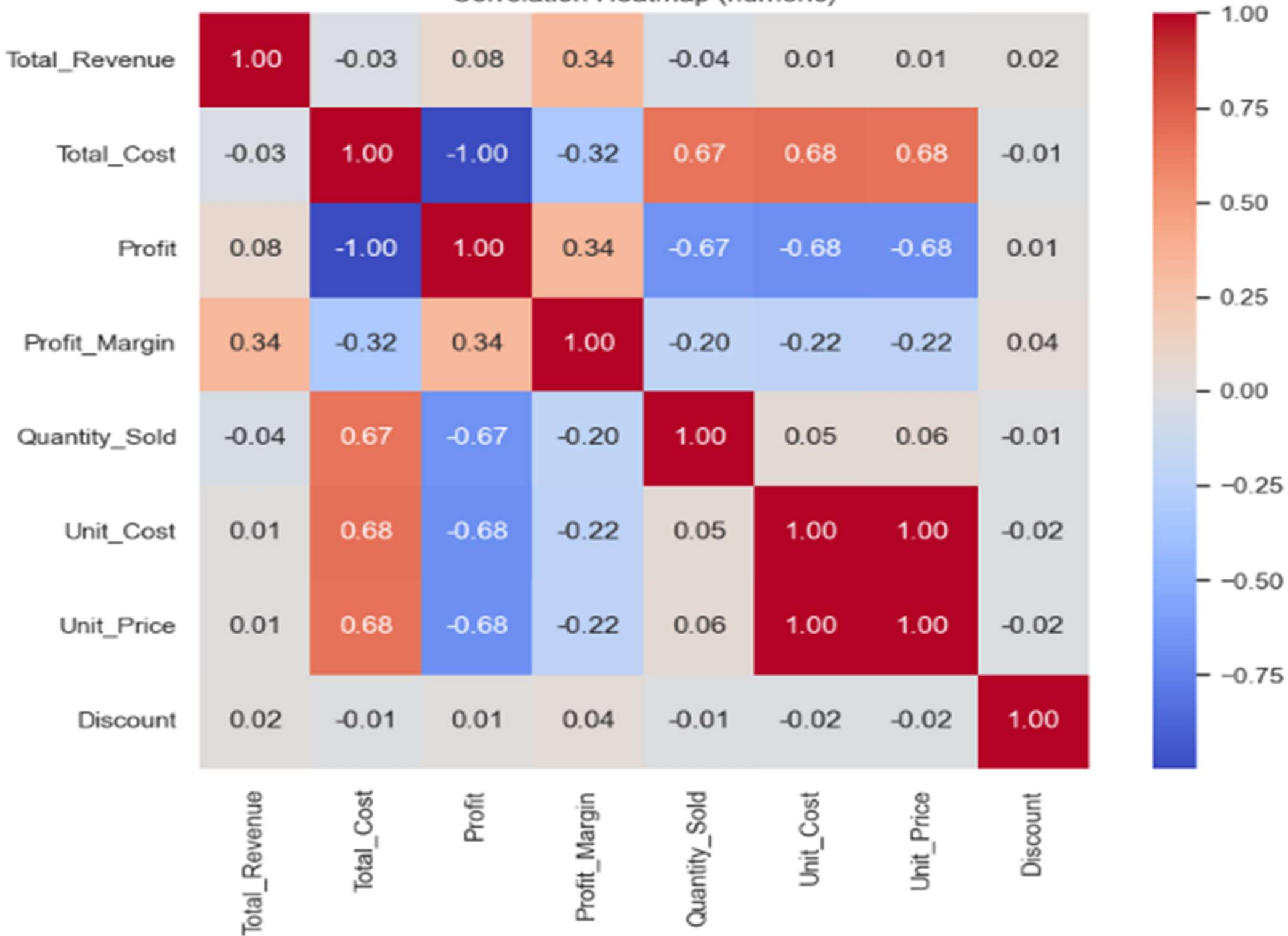
- **Product Strategy Insight**

Occasional high-quantity spikes may come from **specific product categories**, offering opportunities for bundle offers or targeted promotions.

- **Sales Optimization Insight**

A uniform distribution with few bulk buyers means the business is **not overly dependent on large orders**, supporting stable and reliable revenue.

Correlation Heatmap (numeric)



CORRELATION HEATMAP



Profitability is driven more by **cost control** than **revenue growth**, as high Unit_Cost and Total_Cost strongly reduce profit.



Since discounts show **little impact on revenue**, pricing and discount strategies can be optimized without risking major sales loss.



Management should prioritize **cost reduction**, **supplier negotiation**, and **pricing refinement** to improve overall profit margins.

Data Analytics

Key Insights

- Revenue is highest in top-performing categories and regions.
- Discounts between 5–20% show strong demand uplift.
- Profitability varies significantly by product category.
- Revenue shows recognizable monthly patterns, indicating seasonality.
- Correlation heatmap shows strong links between price, cost, and revenue.

Recommendations

- Prioritize high-profit categories for marketing focus.

- Strengthen operations in high-revenue regions.

- Use 5–20% discount range strategically to maximize revenue.

- Monitor low-profit regions/products for corrective action.

- Continue monthly tracking to optimize seasonal inventory.

Thank
You

